Group Project 1: Predictive Statistical Problems Assignment Part 2

Andre Ricardo Caralli Junior,

Bruno de Arantes Leite Sassi,

Luciana Alves de Queiroz Popa, and

Tiago Caramello Ceolato.

Master of Data Analytics, University of Niagara Falls Canada

Predictive Analytics (DAMO-510-1)

Dr. Zeeshan Ahmad

June 13th, 2025

1. Statement of Purpose

Lifestyle choices are a significant determinant of long-term health, contributing to approximately 60% of an individual's overall health and quality of life (Farhud, 2015). Unhealthy habits, such as smoking, excessive alcohol consumption, a poor diet, and a sedentary lifestyle, are primary contributors to chronic conditions like cardiovascular disease, obesity, and metabolic issues.

The goal of this project is to create a predictive model that estimates the likelihood of individuals experiencing health issues. This model will use data from addictive behaviors, such as smoking and drinking, as well as lifestyle factors, including exercise, diet, sleep, and mental health.

The goal of this project is to create a predictive model that estimates the likelihood of individuals experiencing health issues. This model will use data from addictive behaviors, such as smoking and drinking, as well as lifestyle factors, including exercise, diet, sleep, and mental health. Machine learning techniques can identify the relationship between these habits and health outcomes, providing valuable insights for public health interventions aimed at reducing addiction and improving health.

Leveraging a dataset of 3,000 individuals with addiction patterns will allow us to uncover predictive factors for health issues and explore strategies to support public health policies and prevention initiatives.

2. Scope of the Project

The goal of this project is to build a predictive model that estimates the likelihood of an individual experiencing health issues based on their addictive behaviors and other demographic data. The dataset includes 3,000 individuals exhibiting smoking and drinking addiction patterns, as well as key lifestyle features such as age, exercise frequency, diet quality, hours of sleep, BMI, and mental health status.

The project involves data preparation (cleaning and transformation), exploratory data analysis (EDA), and applying machine learning models, such as logistic regression and classification trees, to predict the binary variable ‘*has_health_issues*’. The goal is not to provide medical diagnoses, but rather to predict health risks based on behaviors and provide insights for policymaking and prevention strategies.

3. Background Research and Literature

When it comes to smoking and drinking, several studies are available to inform the links between them and long-term health issues. According to the World Health Organization (WHO, 2023), tobacco is responsible for more than 8 million deaths a year, including the non-smokers – people who inhale smoke because they are exposed to it. Another key insight provided by WHO (2023) is that in 2020 36.7% of tobacco users were man, while a minority of 7.8% were women. Other data that highlights the public importance of studies regarding smoking is that it is highly responsible for diseases such as lung cancer, chronic respiratory disease, and tuberculosis, but beyond that another alarming statistic informed by WHO (2019) was that more than 60,000 children under 5 years old die due to second-hand smoke.

Alcohol addiction is also very harmful with 2.6 million deaths in 2019 related to alcohol consumption, with 2 million being among men (WHO, 2024). Apart from that, the direct diseases caused by alcohol addiction are liver diseases, heart diseases, mental health and behavioral conditions, as well as many cancer types, such as breast, liver, head, and neck cancer. Additionally, according to WHO (2024), young people are more affected by alcohol addiction in general with people between 20 and 39 years being 13% of the alcohol-attributable deaths.

Furthermore, the co-occurring usage of both smoking and alcohol raises concerns, since around 20% of smokers also drink heavily, which is linked with worse morbidity and mortality, but has no current treatment or best practices to treat both addictions together (King & Fucito, 2021).

Given all the issues and concerns brought by cigarettes and alcohol addiction, society and public health are negatively affected by those habits, and studies related to this field are more than welcome to understand how governments can make policies better address these issues. Predictive modeling, particularly machine learning approaches, can offer early insights into at-risk populations and inform effective public health strategies to reduce addiction and its societal impact.

4. Methodology/Strategies

This project employs predictive analytics to explore how addiction-related behaviors and lifestyle factors impact health outcomes. The analysis leverages a structured dataset from Kaggle, featuring data from individuals with patterns of addiction. The primary goals of this research are to examine key lifestyle, demographic, and addiction-related variables and to develop a logistic regression model for estimating the likelihood of health issues. The findings aim to inform public health strategies by identifying behaviors most predictive of adverse health outcomes.

4.1 Dataset Overview

The dataset used in this project was obtained from Kaggle and provided by Yadav (n.d.). It consists of data collected from 3,000 individuals who exhibit addiction behaviors related to smoking and drinking, as well as various demographic and lifestyle factors. These factors include age, employment status, income, mental health, exercise habits, diet quality, sleep duration, and social support. This dataset is ideal for building a predictive model aimed at understanding the relationship between lifestyle choices and health outcomes.

Key Variables:

The dataset contains several key variables, categorized into predicted and explanatory variables:

- Predicted Variable:
  - has_health_issues - A binary variable indicating whether the individual reports having health issues (Yes/No).
- Explanatory Variables:
  - Addiction-related: smokes_per_day, drinks_per_week, age_started_smoking, age_started_drinking, attempts_to_quit_smoking, attempts_to_quit_drinking
  - Lifestyle and Health: exercise_frequency, diet_quality, sleep_hours, bmi, mental_health_status, social_support, therapy_history
  - Demographic-related: age, gender, education_level, employment_status, annual_income_usd, marital_status, children_count, country

4.2 Data Cleaning and Preprocessing

The cleaning and preprocessing phase was essential to prepare the data for analysis. This phase involved several key steps to ensure the dataset was reliable and ready for model development.

I. Removing Irrelevant Columns:

Several columns in the dataset, such as id, name, and city, were removed as they did not contribute useful information for the analysis. These columns are typically used for identification purposes but do not aid in predicting health issues.

II. Handling Missing Data:

The dataset contained missing values represented as 'None' entries in several columns. These 'None' entries were replaced with ‘Never’ to facilitate processing. The missing values were then analyzed and will be addressed in the Feature Engineering phase. The columns with missing values included:

- education_level: 420 missing entries
- social_support: 753 missing entries
- therapy_history: 1014 missing entries

These missing values will be imputed or handled appropriately based on the importance of each feature in the predictive modeling process.

III. Data Type Validation:

It was crucial to ensure that each feature in the dataset was assigned the correct data type for further analysis. The dataset contained a mix of:

- Numerical: Integer (int64), Float (float64)
- Categorical: String (object), representing factors like gender, education_level, and mental_health_status
- Boolean: For the target variable has_health_issues, which indicates the presence of health issues (Yes/No).

These data types were validated, ensuring that categorical variables were appropriately encoded for machine learning models.

The preprocessing steps outlined ensure that the data is clean, well-structured, and appropriate for predictive modeling using techniques such as Logistic Regression and Classification Trees.

4.3 Data Preparation

For feature transformation, categorical variables, such as gender and employment status, were encoded using either label encoding or one-hot encoding to ensure compatibility with machine learning algorithms. Additionally, log transformations were applied to features exhibiting skewed distributions, such as income and BMI, to normalize the data and mitigate the influence of outliers.

Feature selection was carried out using a correlation matrix. A heatmap was generated to visually inspect which variables were most strongly associated with the binary target variable, *has_health_issues* (Yes/No). The most strongly correlated explanatory variables were retained for modeling. These included smoking behavior (smokes_per_day) and drinking behavior (drinks_per_week), as well as lifestyle factors like exercise frequency, diet quality, and sleep hours. Health-related variables such as BMI, mental health status, and social support were also included.

4.4 Model Development Methodology

Logistic regression was selected as the machine learning algorithm for model development due to the binary nature of the target variable (*has_health_issues*: Yes/No). The model estimates the probability P(Y=1|X)P(Y = 1 \mid X)P(Y=1|X), where YYY represents the presence of health issues, and XXX is the vector of explanatory variables.

The model development process began with dataset loading and preprocessing. The data was loaded using the Pandas library, and categorical variables were encoded and transformed as necessary. For exploratory data analysis (EDA – Appendix A), summary statistics were computed, and visualizations such as boxplots and histograms were generated to examine the data distributions and detect outliers.

In the model training phase, a train-test split was performed, typically using a 70/30 or 80/20 ratio, to prevent overfitting and allow for the evaluation of model performance on unseen data. Logistic regression was then applied using the *sklearn.linear_model.LogisticRegression* module.

Model evaluation involved the use of accuracy, precision, recall, and confusion matrix metrics to assess the model's performance. The coefficients of the logistic regression model were examined to interpret the direction and strength of each predictor's influence on the likelihood of health issues.

4.5 Business and Public Health Impact

By identifying key variables that contribute to health issues among individuals with addiction patterns, this project provides a data-driven foundation for public health policy-making. It offers predictive insights that can aid healthcare professionals in targeting lifestyle interventions more effectively. Furthermore, the findings support preventive health strategies for governments and NGOs by uncovering actionable variables, such as poor sleep, lack of exercise, and high substance consumption, which can be addressed to improve overall public health outcomes.

5 Business Impact and Conclusion

The present project demonstrates the potential of predictive analysis in public health by identifying the behavioral and demographic aspects which were self-reported by each respondent. Through the logistic regression model a moderate predictive performance was present and key variables were identified, such as the smoking frequency, alcohol consumption, sleep duration, BMI, and mental health status, that significantly correlate with the health problems, as can be observed on Figure 1.

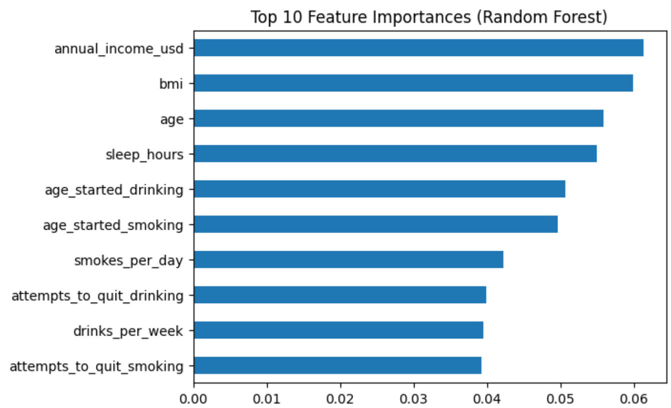


Figure 1. Top 10 Feature Importances (Random Forest).

In addition to logistic regression, the Random Forest was also applied to compare predictive performance. However, both models resulted in low accuracy and area under the curve scores (AUC), approximately 49% for logistic regression and 51% for Random Forest, which indicates that the models were not significantly better than random guessing, as shown in the Figure 2.

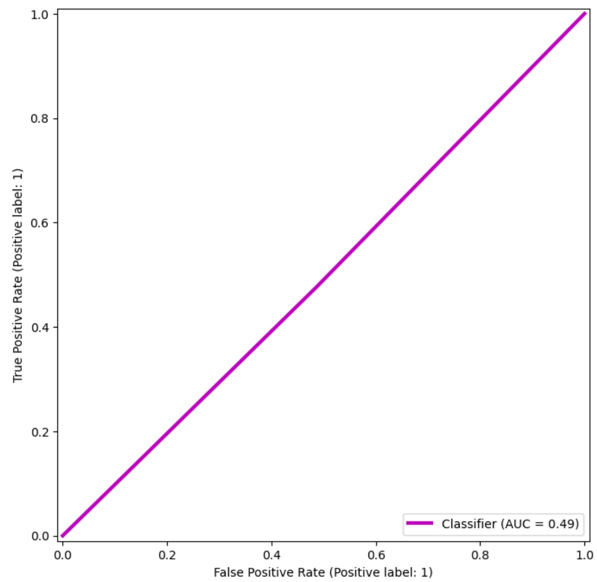


Figure 2. ROC Curve Plot.

The Confusion Matrix offers even more insights into how our model performed by detailing the correct and incorrect classifications. When looking at Figure 3, it is possible to see the Random Forest model misclassified both positive and negative cases at similar rates. This reinforces our interpretation that the classifier is performing close to random chance, aligning with the low predictive value observed in its accuracy and AUC scores.

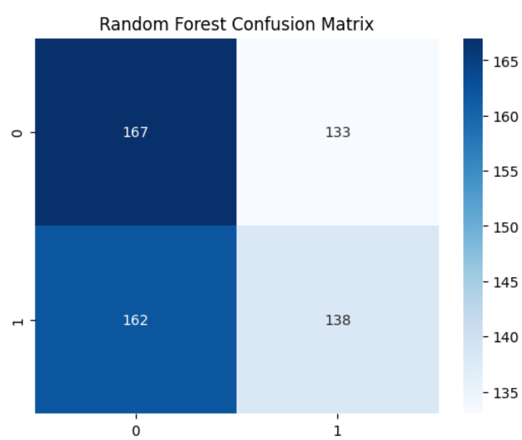


Figure 3. Random Forest Confusion Matrix.

From a perspective of a business and policy-making context, these findings can directly inform how healthcare providers, NGOs, and government agencies design targeted intervention programs. For example, community-level campaigns which promotes regular exercise, better diet, and reduced substance usage could be prioritized in regions or demographics that our model identifies as high-risk. Additionally, employers and insurance companies might leverage similar models in order to implement preventive wellness initiatives, potentially reducing long-term healthcare costs and absenteeism.

The predictive insights observed from this model also enable a proactive approach to public health. Instead of simply reacting to chronic illness after its onset, stakeholders can use these forecasts to deliver early interventions and resources to vulnerable populations. Furthermore, the established correlation between unemployment, low income and health issues highlights the connection between social determinants and health, emphasizing the need for cross-sector strategies that address both lifestyle and socioeconomic conditions.

Nevertheless, as in every study, our observations present limitations. First, our dataset is synthetic, which might not perfectly reflect the real-world population information. Also, some variables had a lot of missing values, that could be better handled by the utilization of more advanced techniques such as Gradient Boosting. Furthermore, the model evaluation metrics such as AUC, precision, recall, and F1-score remained close to 0.5, highlighting the difficulty in making reliable predictions with synthetic data. In addition, the binary outcome variable simplifies the complex health dynamics into a single indicator, which may not capture all nuances.

It is recommended that future works incorporate time-series data to capture health trends over time, using real-world data for model accuracy, or even developing tailored models for specific age or income groups. Despite the limitations, the project can illustrate how data-driven approaches can aid public health strategies based on reliable data.

References

Farhud, D. D. (2015, November). *Impact of Lifestyle on Health*. Iranian Journal of Public Health. https://pmc.ncbi.nlm.nih.gov/articles/PMC4703222/#B1

Yadav, K. (n.d.). *Cigarettes & Alcohol Addiction*. Kaggle. https://www.kaggle.com/datasets/khushikyad001/cigarettes-and-alcohol-addiction/data

King, A., & Fucito, L. (2021, September 13). *Cigarette smoking and heavy alcohol drinking: The challenges and opportunities for combination treatments | American Journal of Psychiatry*. Psychiatry Online. https://psychiatryonline.org/doi/10.1176/appi.ajp.2021.21070692

World Health Organization. (2019, May 29). *Who highlights huge scale of tobacco-related lung disease deaths*. World Health Organization. https://www.who.int/news/item/29-05-2019-who-highlights-huge-scale-of-tobacco-related-lung-disease-deaths?utm_source=chatgpt.com

World Health Organization. (2023, July 31). *Tobacco*. World Health Organization. https://www.who.int/en/news-room/fact-sheets/detail/tobacco?utm_source=chatgpt.com

World Health Organization. (2024, June 28). *Alcohol*. World Health Organization. https://www.who.int/news-room/fact-sheets/detail/alcohol

Appendix A

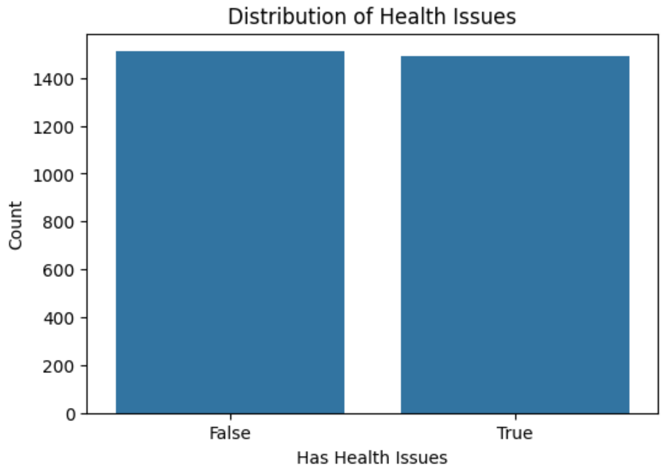
This appendix provides an Exploratory Data Analysis (EDA) into the dataset used in this project. The dataset offers insights into various demographic, lifestyle, and health-related variables, including addiction behaviors, mental health status, exercise habits, diet quality, and more. The main goal of this Exploratory Data Analysis (EDA) is to explore how health issues (i.e., whether an individual reports having health issues or not) correlate with different behavioral and demographic factors. The section below presents key patterns and trends observed in the dataset, supported by visualizations.

Key Insights from the EDA

a. Health Issues Distribution

The bar chart in Figure 1 presents the distribution of individuals with and without health issues. The "False" category, indicating individuals without health issues, slightly outnumbers the "True" category, suggesting a near 50/50 distribution between individuals with and without health issues. The small difference between the two groups should be interpreted with caution, as this could indicate an approximately balanced health status across the dataset, but further analysis is needed to determine whether this distribution is statistically significant.

Figure A1. Distribution of health issues.

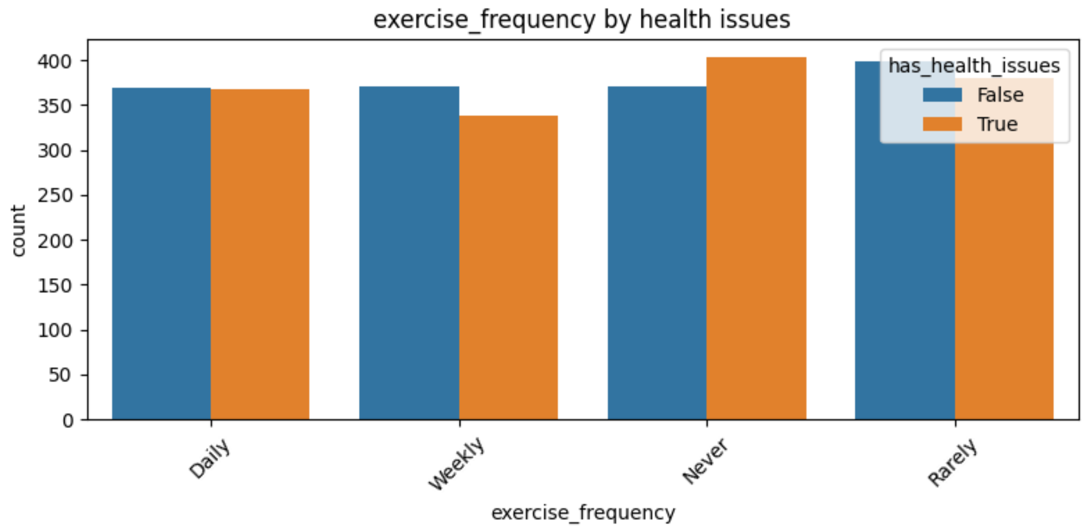


Note. This bar chart presents the distribution of individuals with and without health issues.

b. **Exercise Frequency and Health Issues**

The following graph (Figure 2) shows the exercise frequency distribution for individuals with and without health issues. While the data shows a higher frequency of exercise among individuals without health issues, the differences are modest, with both groups reporting varying exercise habits. The findings suggest that individuals without health issues may exercise more often, but the exact strength of this association requires further analysis, such as statistical testing to determine significance.

Figure A2. Exercise frequency by health issues.

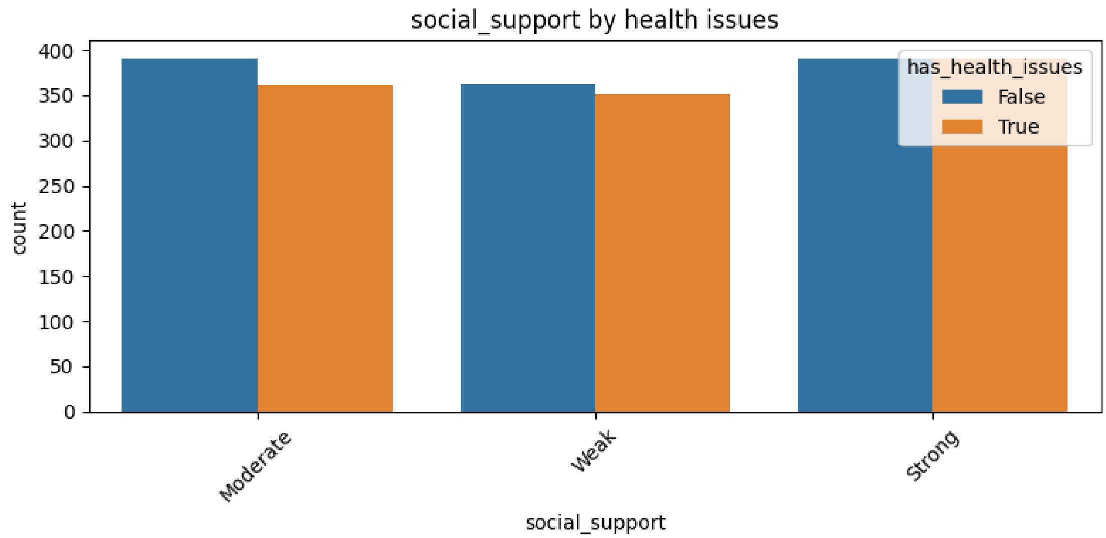


Note. This graph illustrates the exercise frequency distribution for individuals with and without health issues.

c. **Social Support and Health Issues**

The second graph (Figure 3) compares social support levels between individuals with and without health issues. The data shows a slight difference, with individuals without health issues reporting slightly stronger social support. However, most individuals in both groups report moderate to strong social support, and the differences are not substantial enough to make definitive conclusions about the impact of health issues on social support levels.

Figure A3. Social support by health issues.

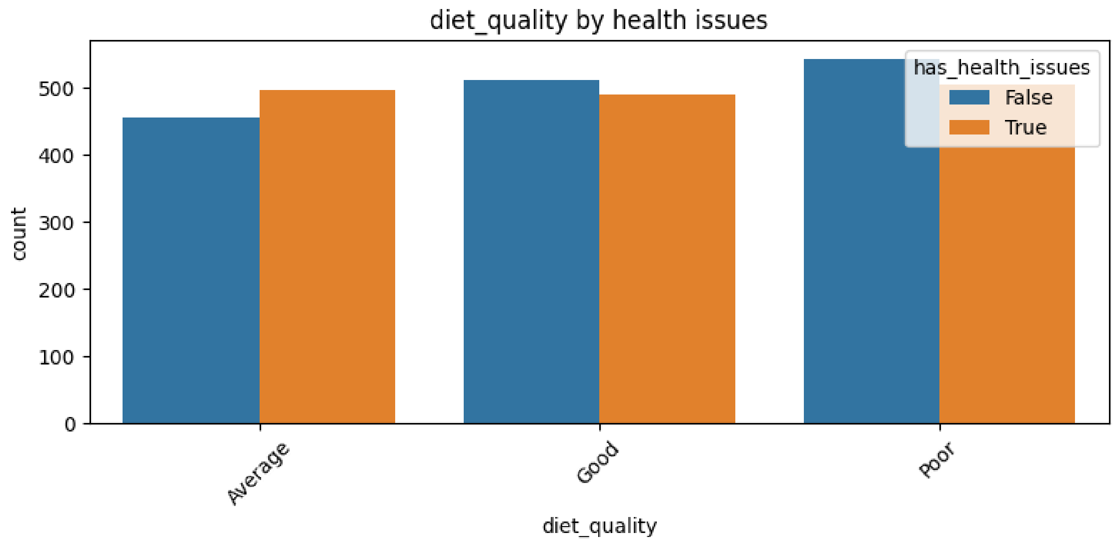


Note. This bar chart depicts the level of social support among individuals with and without health issues.

d. **Diet Quality and Health Issues**

The third graph (Figure 4) compares diet quality between individuals with and without health issues. It reveals a somewhat higher proportion of individuals with health issues reporting "Poor" diet quality. However, the difference is not significant enough to suggest a direct correlation between health status and diet quality. The distribution remains relatively balanced, and more analysis is needed to confirm any potential link.

Figure A4. Diet quality by health issues.

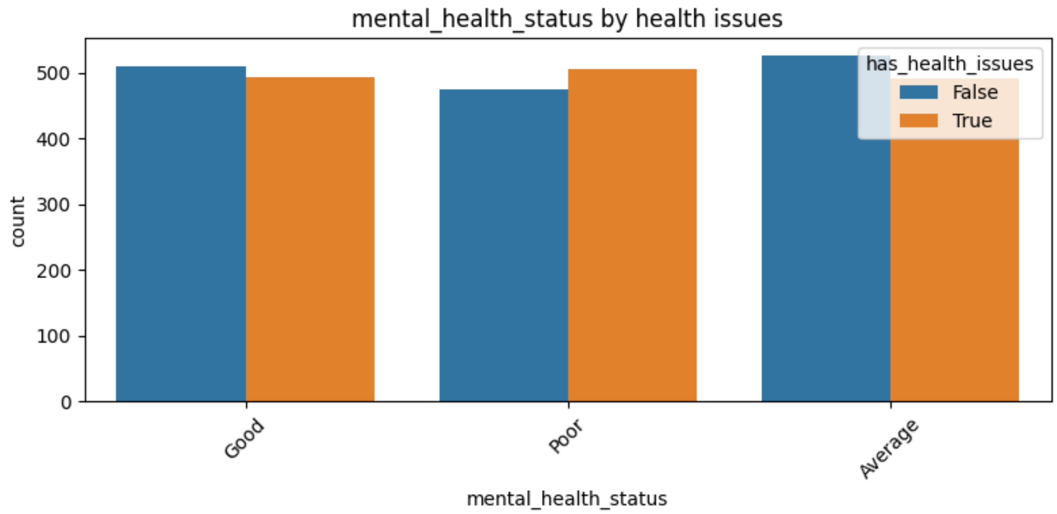


Note. The bar chart in this figure compares diet quality between individuals with and without health issues.

e. **Mental Health Status by Health Issues**

The fourth graph (Figure 5) shows the mental health status distribution for individuals with and without health issues. Individuals with health issues tend to report poorer mental health more frequently than those without health issues. While this trend is apparent, the data does not provide conclusive evidence that health issues directly cause poorer mental health, as other factors may also contribute to mental health status.

Figure A5. Mental health status by health issues.

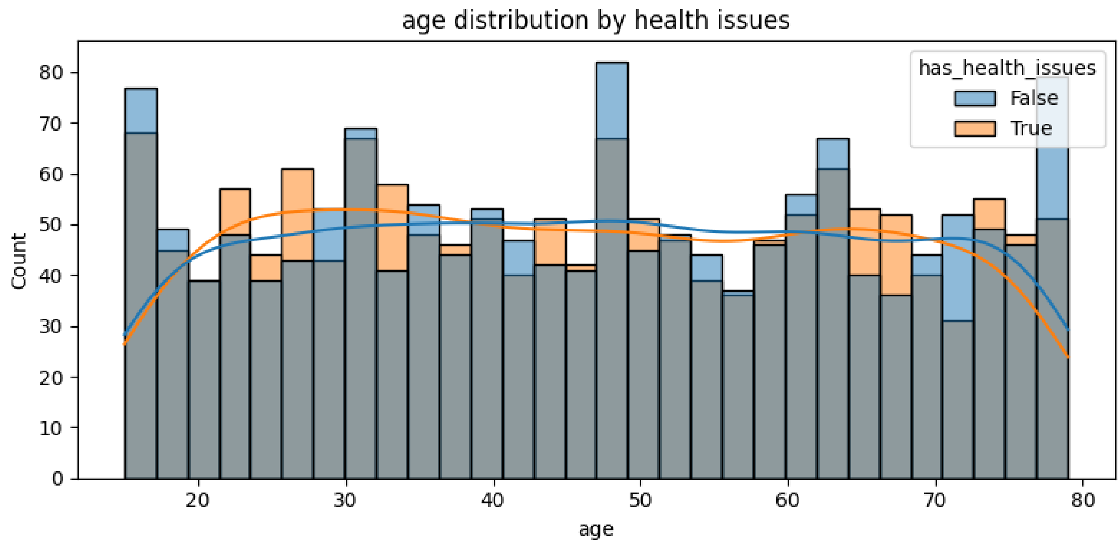


Note. This graph compares mental health status between individuals with and without health issues.

f. **Age Distribution by Health Issues**

The distribution of age by health issues shows slight variations between individuals with and without health issues. While the age distribution is similar for both groups, there are minor differences that may warrant further investigation. The graph does not suggest a strong relationship between age and the likelihood of reporting health issues, but this observation could be further explored through statistical testing to confirm its significance.

Figure A6. Age Distribution by Health Issues

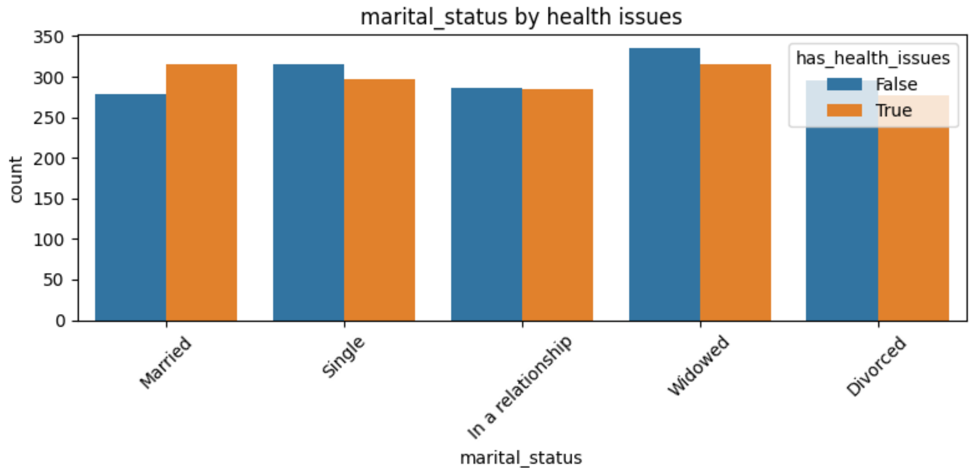


Note. This distribution chart shows the age breakdown of individuals with and without health issues.

g. **Demographic Factors**

Variables such as gender, employment status, marital status, and education level reveal key trends based on health issues. For example, individuals without health issues tend to have better employment status and more stable marital relationships. To deepen this analysis, we can consider the distribution of marital status based on health issues (Figure 7).

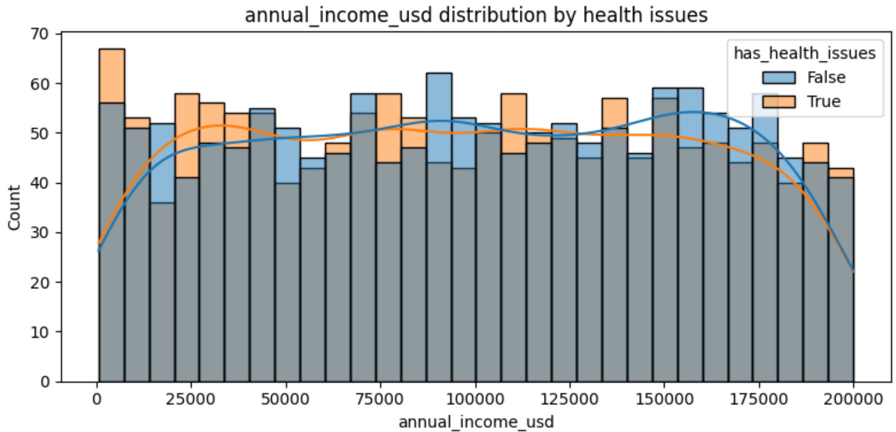
Figure A7. Marital Status by Health Issues



Note. The graph indicates that individuals with health issues are slightly less likely to be married or in a stable relationship compared to those without health issues. This observation could suggest that health problems may affect personal relationships, potentially due to the strain health issues place on partnerships.

Additionally, income levels could be explored as another demographic factor. The distribution of income by health issues (Figure 8) suggests that individuals without health issues tend to occupy higher income brackets, highlighting the potential connection between better health and improved financial outcomes. This reinforces the notion that health may influence employment opportunities and economic stability.

Figure A8. Income distribution by health issues

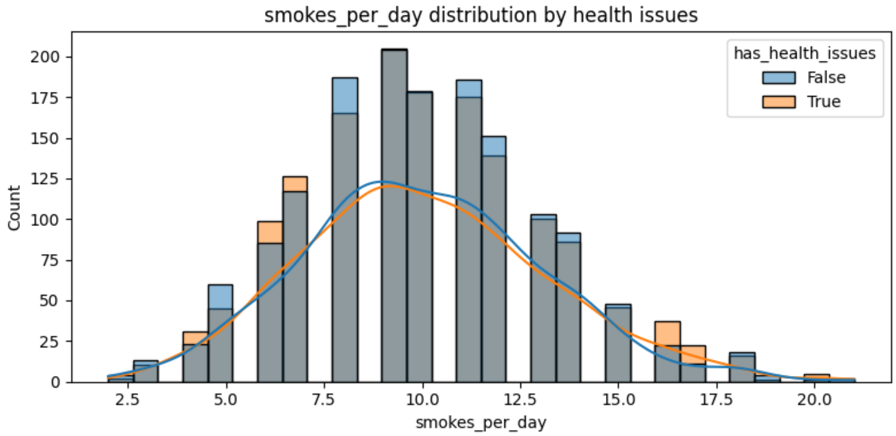


Note. The graph reveals that individuals without health issues have a higher presence in higher income ranges, which could reflect the increased access to resources and better job stability.

h. **Smoking and Drinking Behavior**

The distribution of smoking behavior shows a higher prevalence of smoking among individuals with health issues. This aligns with existing literature suggesting that health problems are often correlated with higher rates of smoking, potentially as individuals cope with stress or other underlying factors. The smoking behavior distribution (Figure 9) clearly demonstrates that those with health issues tend to smoke more frequently, with higher counts observed in the 7-10 cigarettes per day range.

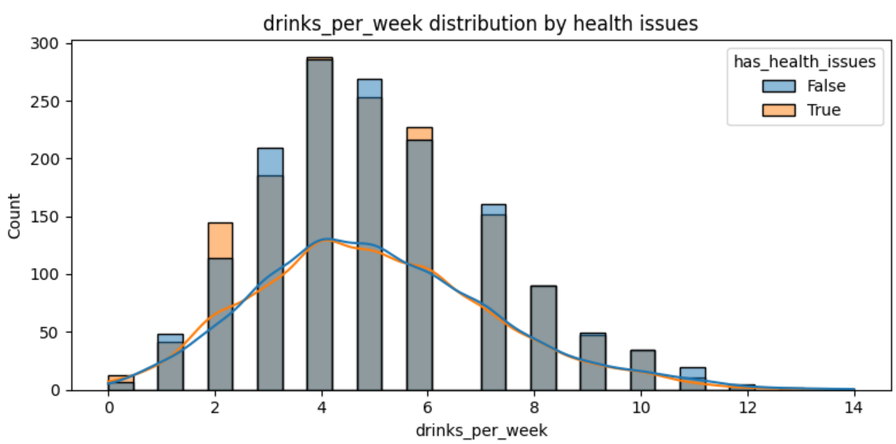
Figure A9. Smoking Behavior by Health Issues



Note. The graph shows that individuals with health issues are more likely to smoke daily or in higher quantities compared to those without health issues. This visual reinforces the narrative that smoking is a significant behavioral factor impacting health.

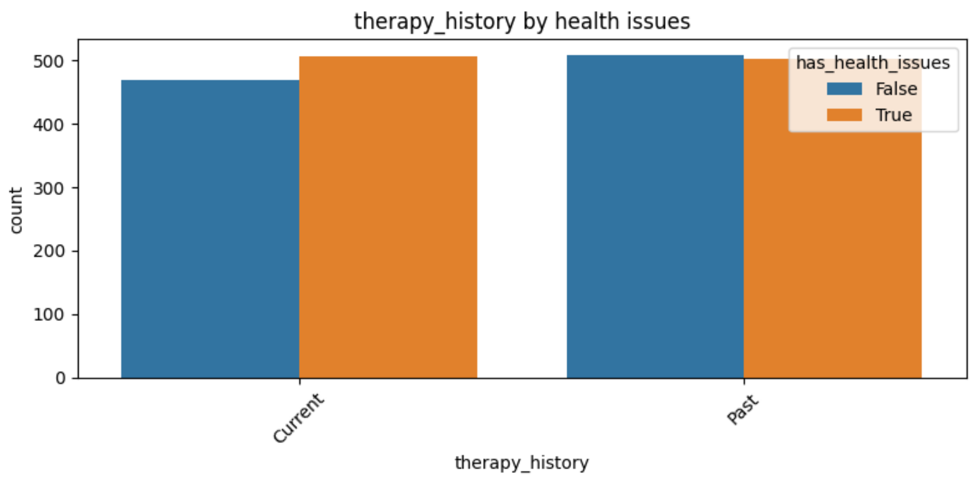
On the other hand, while the drinking behavior distribution (Figure 10) shows no strong pattern overall, individuals with health issues tend to drink more frequently in the lower consumption ranges. This could be reflective of a coping mechanism or a dependency in certain cases.

Figure A10. Drinking Behavior by Health Issues



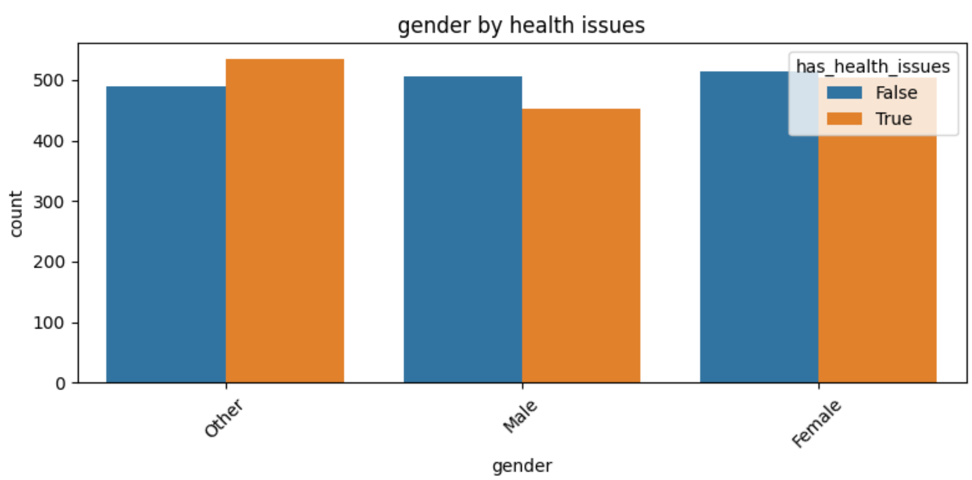
Note. The graph highlights that while drinking habits vary, individuals with health issues seem to drink more frequently in the lower consumption ranges, suggesting that alcohol consumption may serve as a coping mechanism for some.

Figure A11. Therapy History by Health Issues



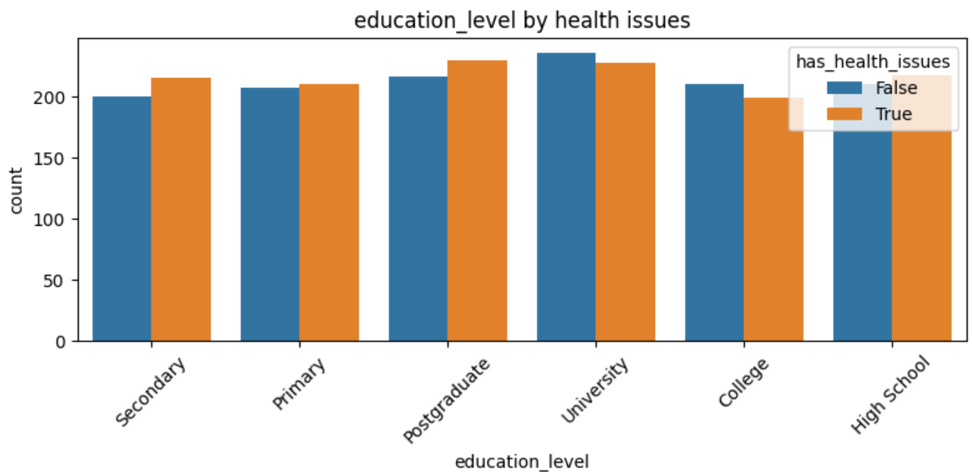
Note. This graph compares the therapy history of individuals with and without health issues. It shows that individuals with health issues tend to report a higher proportion of having a past or current therapy history, but the differences between the two groups are not substantial. Further analysis could explore whether this relationship is statistically significant.

Figure A12. Gender by Health Issues



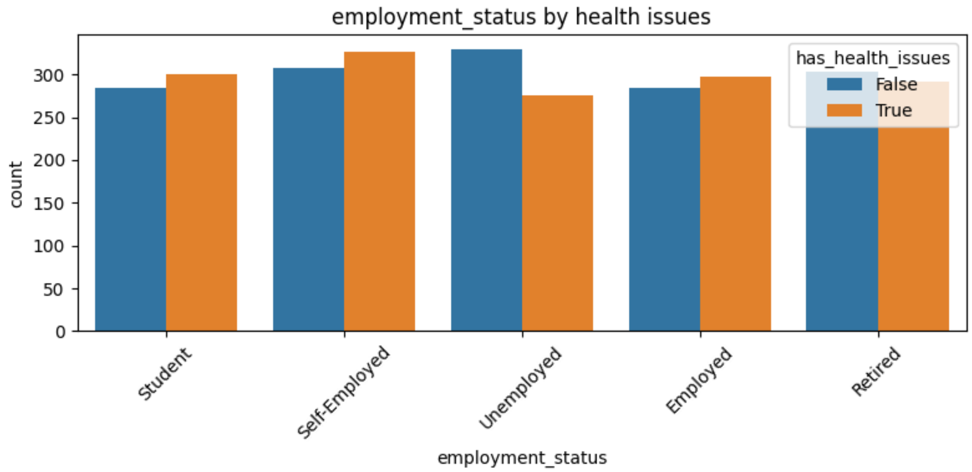
Note. This chart shows the distribution of gender among individuals with and without health issues. While there is a slightly higher proportion of males in both groups, the gender distribution is similar across individuals with and without health issues. Gender does not appear to be a key differentiator between the two groups in this dataset.

Figure A13. Education Level by Health Issues



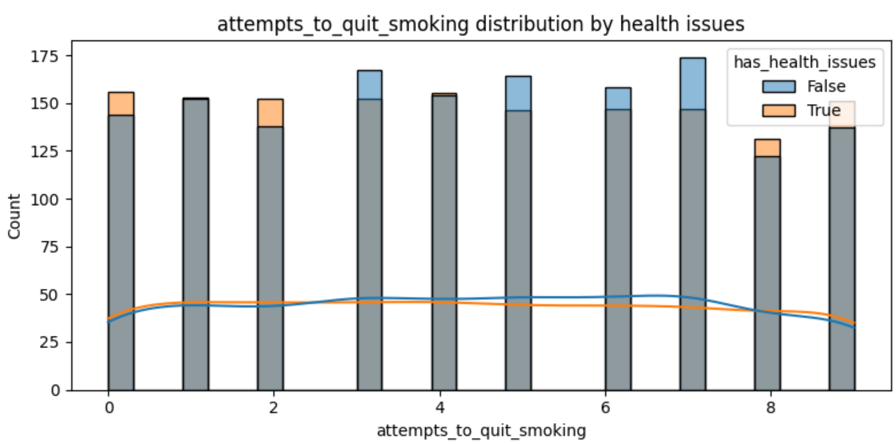
Note. This figure presents the education level of individuals with and without health issues. The distribution suggests that most individuals, regardless of health status, have attained at least some higher education, with no substantial differences between the two groups. It appears that education level does not strongly correlate with the presence of health issues in this dataset.

Figure A14. Employment Status by Health Issues



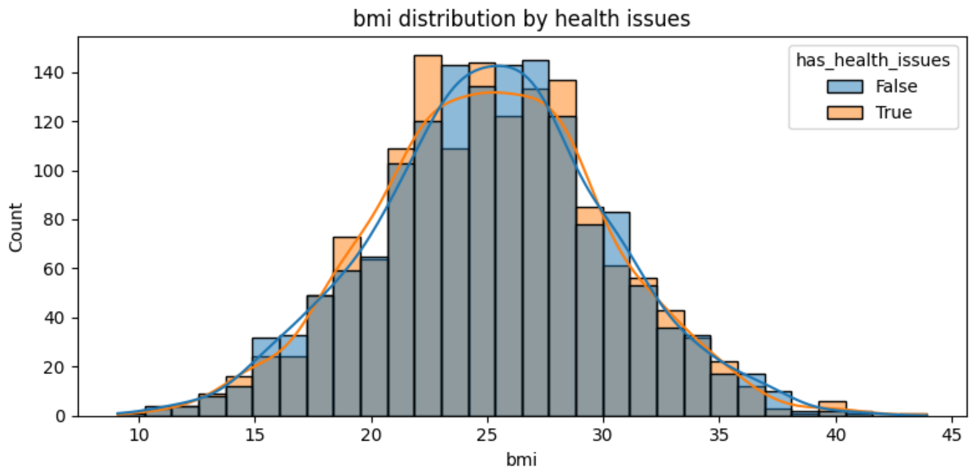
Note. This chart compares the employment status of individuals with and without health issues. Individuals with health issues show a higher percentage of unemployment and part-time employment compared to those without health issues, who are more likely to be employed full-time. This suggests a potential relationship between employment status and health, although more detailed statistical analysis is needed to confirm this trend.

Figure A15. Attempt to Quit Smoking/Drinking Distribution by Health Issues



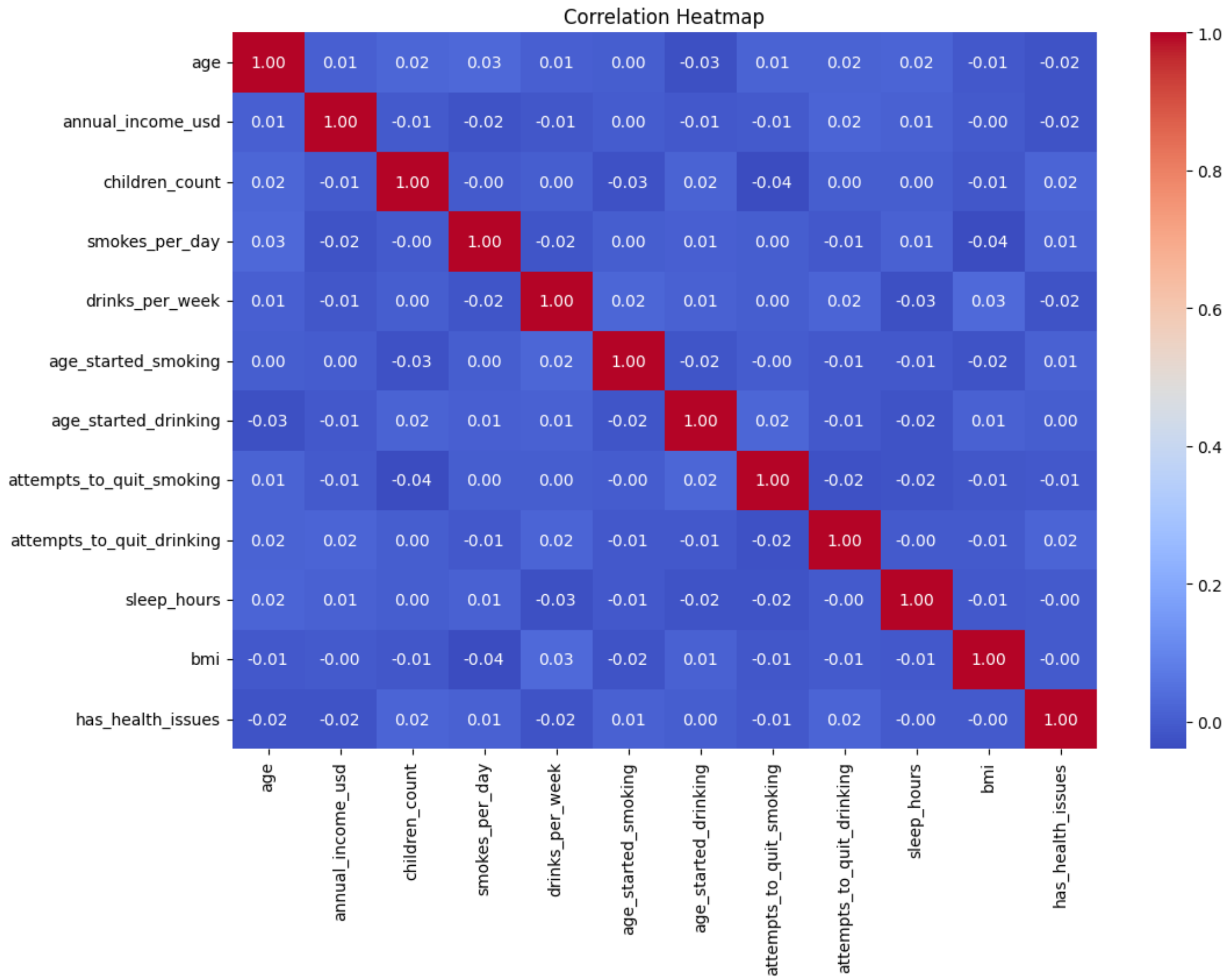
Note. This figure compares the number of attempts to quit smoking and drinking between individuals with and without health issues. The data indicates that individuals with health issues report slightly more attempts to quit smoking and drinking. However, the frequency of attempts is relatively spread out, and no clear pattern emerges that strongly links health issues to quitting behavior.

Figure A16. BMI, Sleep Hours, and Other Distributions by Health Issues



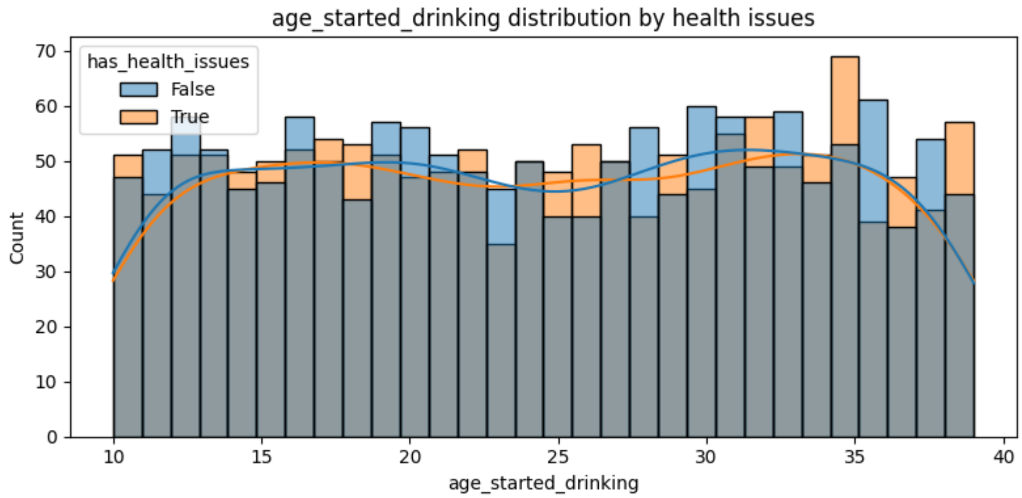
Note. This set of distributions compares BMI, sleep hours, and other health-related factors between individuals with and without health issues. The data shows that individuals with health issues tend to have slightly higher BMIs and report fewer hours of sleep, but these differences are not large enough to draw definitive conclusions without further analysis. It suggests that lifestyle factors like sleep and BMI could be relevant but requires deeper investigation.

Figure A17. Correlation Heatmap



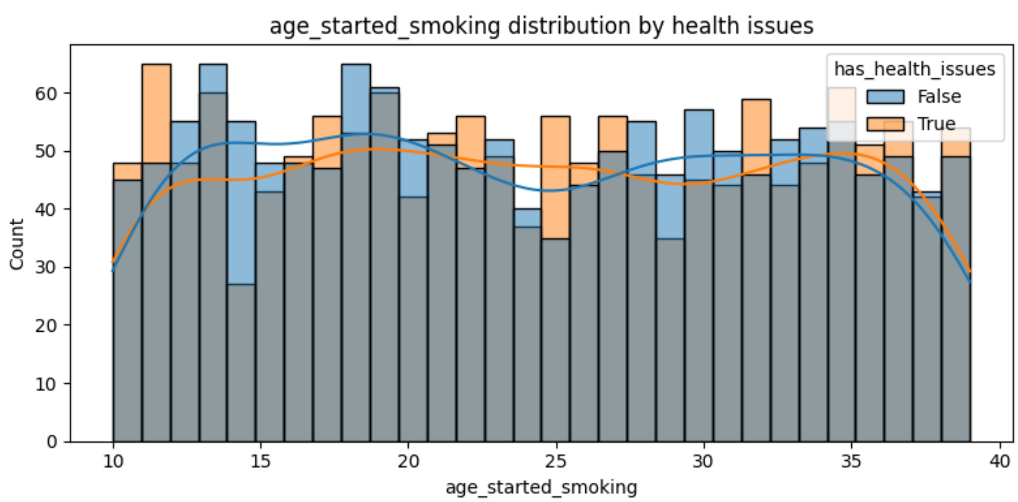
Note. The correlation heatmap displays relationships between variables such as smoking, drinking, and health issues. It reveals moderate correlations between "smokes_per_day" and "drinks_per_week," suggesting a potential relationship between smoking and drinking behaviors. The correlation between age and health issues is weak, indicating that age does not have a strong influence on whether someone reports having health issues.

Figure A18. Age Started Drinking Distribution by Health Issues



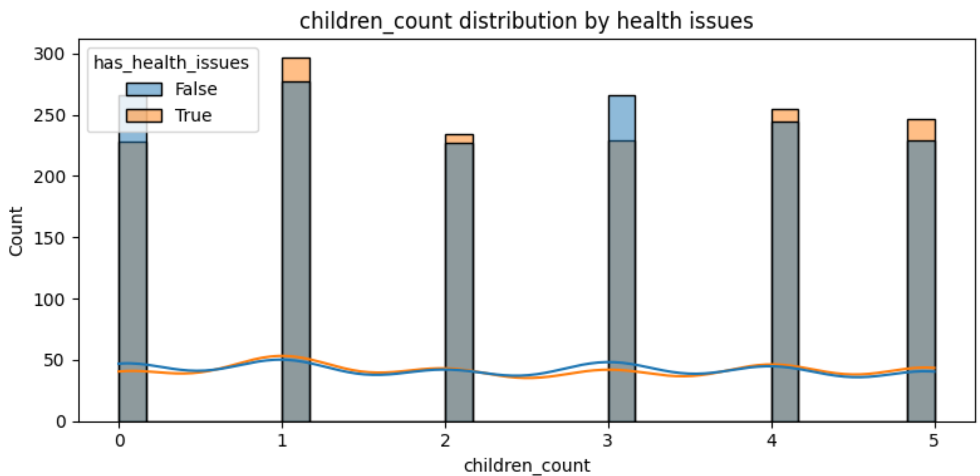
Note. The graph presents the distribution of individuals' ages when they first started drinking, segmented by whether they report having health issues (True or False). The bar chart reveals a relatively uniform distribution across different ages, with a slightly higher count of individuals starting at ages 18–20. The line plot indicates that those with health issues (orange line) tend to start drinking at an earlier age and have a more consistent distribution across the age groups compared to those without health issues (blue line).

Figure A19. Age Started Smoking Distribution by Health Issues



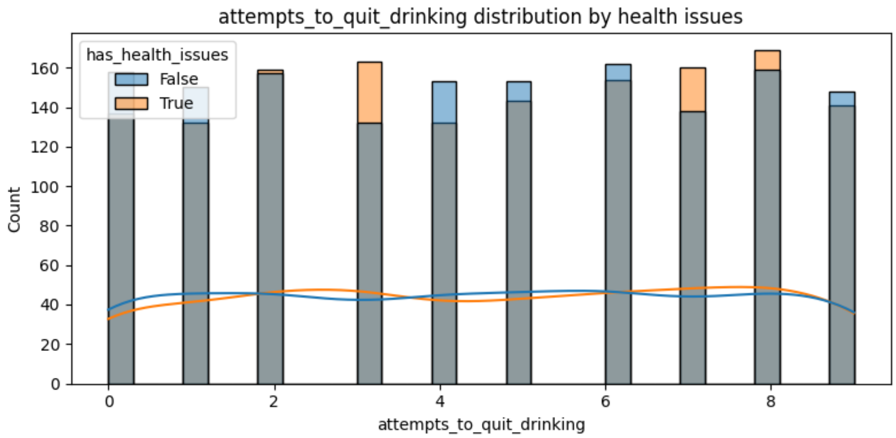
Note. This graph shows the distribution of ages at which individuals first started smoking, differentiated by their health issue status. A similar pattern is observed as with drinking, where the individuals with health issues (orange line) show a more evenly distributed range across the ages, with slightly more individuals starting to smoke between the ages of 15 and 20. The health issues variable appears to have a moderate influence on the age at which smoking starts.

Figure A20. Children Count Distribution by Health Issues



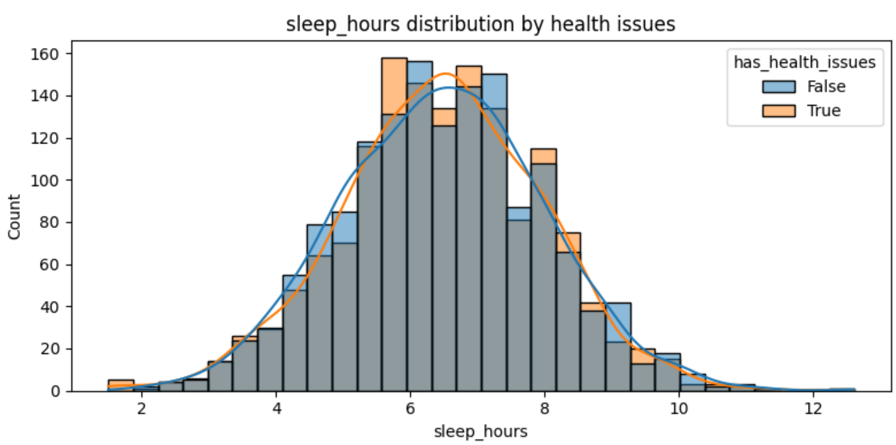
Note. The graph illustrates the count of children per individual and its relationship with having health issues. The bar chart demonstrates that most individuals have between 0 and 1 child, with a notably higher count of individuals reporting no health issues (blue). The distribution is consistent across different numbers of children, with the health issue group (orange) showing a slight increase in the number of individuals reporting higher children counts (3 and 4).

Figure A21. Attempts to Quit Drinking Distribution by Health Issues



Note. This graph compares the number of attempts individuals made to quit drinking and whether they have health issues. The distribution of attempts appears consistent for both groups (those with and without health issues), with the majority of individuals making 1–2 attempts. The line graph, however, shows that individuals with health issues (orange line) report a somewhat higher number of attempts overall compared to individuals without health issues (blue line), indicating a potentially stronger desire to quit drinking among the health issue group.

Figure A22. Sleep Hours Distribution by Health Issues



Note. The graph shows the distribution of individuals' reported sleep hours per night, categorized by whether they have health issues. The histogram reveals that most individuals sleep between 6 and 8 hours, with the distribution skewing slightly towards higher sleep hours for individuals without health issues (blue). The individuals with health issues (orange) show a more even distribution across the sleep hours, with fewer reporting extreme sleep behaviors, indicating a potential connection between health issues and sleep patterns.